

Ex. No: 13 File Organization Techniques

PROGRAM A: SEQUENTIAL FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;

    fp = fopen("student.dat", "w");
    File Organization Techniques

    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);

    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);
    printf("\nRecord Details\n");
    printf("Register Number : %d\n", s.regno);
```

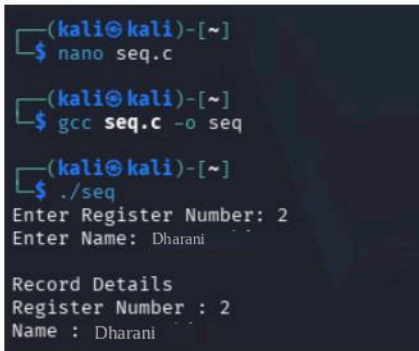
```
printf("Name : %s\n", s.name);

fclose(fp);

return 0;

}
```

Output:



```
(kali㉿kali)-[~]
└─$ nano seq.c

(kali㉿kali)-[~]
└─$ gcc seq.c -o seq

(kali㉿kali)-[~]
└─$ ./seq
Enter Register Number: 2
Enter Name: Dharani

Record Details
Register Number : 2
Name : Dharani
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Register Number:"

read regno

echo "Enter Name:"

read name

echo "$regno $name" > student.txt

echo "Contents of File"

cat student.txt
```

Output:

```
(kali@kali)~$ bash lab13-1.sh
Enter Register Number:
22
Enter Name:
kali
Contents of File
22 kali
```

PROGRAM B: DIRECT (RANDOM) FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;
    fp = fopen("random.dat", "wb+");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fwrite(&s, sizeof(s), 1, fp);
    rewind(fp);
    fread(&s, sizeof(s), 1, fp);
    printf("\nRecord Found\n");
    printf("Reg No : %d\n", s.regno);
```

```
printf("Name : %s\n", s.name);  
  
fclose(fp);  
  
return 0;  
  
}
```

Output:



```
(kali@kali)-[~]  
$ nano Direct.c  
  
(kali@kali)-[~]  
$ gcc Direct.c -o Direct  
  
(kali@kali)-[~]  
$ ./Direct  
Enter Register Number: 22  
Enter Name: Dharani  
  
Record Found  
Reg No : 22  
Name : Dharani
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Record:"  
  
read rec  
  
echo "$rec" > random.txt  
  
echo "Random Access Record"  
  
sed -n '1p' random.txt
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab13-2.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab13-2.sh  
  
(kali㉿kali)-[~]  
$ bash lab13-2.sh  
Enter Record:  
22  
Random Access Record  
22
```

PROGRAM C: INDEXED FILE ORGANIZATION

C PROGRAM

```
#include <stdio.h>
```

```
struct student
```

```
{
```

```
int regno;
```

```
char name[20];
```

```
};
```

```
int main()
```

```
{
```

```
struct student s[3];
```

```
int key, i;
```

```
printf("Enter 3 Student Records\n");
```

```
for(i=0;i<3;i++)
```

```
{
```

```
scanf("%d %s",&s[i].regno,s[i].name);
```

```
}
```

```
printf("Enter Register Number to Search: ");
```

```
scanf("%d",&key);
```

```
for(i=0;i<3;i++)  
{  
if(s[i].regno==key)  
{  
printf("\nRecord Found\n");  
printf("Reg No : %d\n",s[i].regno);
```

76

```
printf("Name : %s\n",s[i].name);  
return 0;  
}  
}  
printf("Record Not Found\n");  
return 0;  
}
```

Output:

```
(kali@kali)-[~]  
$ nano indexed.c  
  
(kali@kali)-[~]  
$ gcc indexed.c -o indexed  
  
(kali@kali)-[~]  
$ ./indexed  
Enter 3 Student Records  
22 Pasupathi  
23 Dhanush  
24 Gokul  
Enter Register Number to Search: 22  
  
Record Found  
Reg No : 22  
Name : Dharani
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Student Records"  
  
echo "101 Arun" > index.txt  
  
echo "102 Kumar" >> index.txt  
  
echo "103 Ravi" >> index.txt  
  
echo "Enter Register Number to Search:"  
  
read key  
  
grep "^$key" index.txt
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab13-3.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab13-3.sh  
  
(kali㉿kali)-[~]  
$ bash lab13-3.sh  
Enter Student Records  
Enter Register Number to Search:  
22
```